Unsupervised Across Domain Consistency-Difference Network for Hyperspectral Image Super-Resolution

Zhiling Guo, Jingwei Xin, Nannan Wang, *Member, IEEE*, Jie Li, Xiaoyu Wang, *Fellow, IEEE*, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— Without reducing the spectral resolution, hyperspectral image (HSI) super-resolution has achieved remarkable progress thanks to the success of deep neural networks. However, existing methods cannot fully excavate the latent high-frequency details only in the single spatial domain. Different from existing methods that only achieves the super-resolution task in spatial domain, we optimize the amplitude spectrum and phase spectrum in frequency domain to obtain high-resolution HSI (HR-HSI). We propose a new unsupervised framework to reconstruct HR-HSI using only the observed low resolution (LR) HSI and HR multispectral image (MSI). Based on triple-level modeling, the encoder–decoder learns abundant features including contextual information from multiple scales. In addition, we propose iterative across domain consistency-difference (ADCD) module, which is embedded between encoder and decoder. In ADCD module, three parallel convolution streams (amplitude spectrum adjustment branch, phase spectrum adjustment branch and spatial domain branch) are used to explore the consistency-difference between each other, which is preserved by memory units within the module. Particularly, we embed the dilated causal convolution in the frequency domain processing branch, which is convenient to flexibly adjust the receptive field and adapt to different domains. Extensive experiments are conducted on widely used datasets in comparison with state-of-the-art models, demonstrating the advantage of the proposed method.**

***Index Terms*— Amplitude spectrum, hyperspectral image (HSI), phase spectrum, super-resolution, unsupervised.**

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Zhiling Guo, Jingwei Xin, and Nannan Wang are with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: zhiling@stu.xidian.edu.cn; jwxin@xidian.edu.cn; nnwang@xidian.edu.cn).

Jie Li is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: leejie@mail.xidian.edu.cn).

Xiaoyu Wang is with the School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: fanghuaxue@gmail.com).

Xinbo Gao is with the Chongqing Key Laboratory of Image Cognition, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: gaoxb@cqupt.edu.cn).

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I. Introduction

HYPERSPECTRAL image (HSI) is one of the important media for human perception and analysis of the world, covering the physical characteristics, material content, and distribution of the objects taken. Benefiting from these characteristics, HSI has been used to enhance the performance of a wide range of computer vision and remote sensing tasks, such as classification [1], [2], environmental monitoring [3], change detection [4], and so on [5]. For an HSI, there is a trade-off between spatial resolution and spectral resolution to ensure that the signal received by the remote sensing imaging unit is strong enough. High spatial resolution and spectral resolution are desirable for specific tasks. Considering the fact that HSI has high spectral resolution and low spatial resolution, whereas multispectral image (MSI) has the opposite, a natural approach is to achieve the high-resolution (HR) HSI by fusing low-resolution (LR) HSI and HR-MSI.

Most previously reported methods [6], [7], [8] generally leverage hand-crafted priors knowledge to regularize the mathematical degradation model between LR-HSI, HR-MSI, and the latent HR-HSI. Then, different optimization strategies are used to achieve the optimal solution. Matrix factorization and spectral unmixing are based on exploring the effective representation of high-dimensional spectral vectors to reconstruct HR-HSI. For example, CNMF [8], a classic method, alternately decomposes LR-HSI and HR-MSI to obtain the corresponding endmember matrix and abundance matrix. The estimated endmember matrix of the LR-HSI and estimated abundance matrix of the HR-MSI are multiplied to reconstruct the HR-HSI. Besides, because the spectral information of HSI is redundant, the super-resolution method based on sparse representation has great attraction. Kawakami et al. [9] used sparse prior to learn spectral bases from LR-HSI, and then performed sparse coding on HR-MSI to estimate coefficient matrix. To further boost reconstruction performance, the papers [10], [11] exploit low-rank prior to adopt for exploring the intrinsic spectral correlation of HSI. In a word, the integration of various handcrafted priors, such as physical spectral mixing, sparsity, similarity, and low-rankness improves the performance of HSI super-resolution, but such priors may not hold in some complicated scenarios.

In contrast to the traditional methods, without exploiting handcrafted priors, deep learning-based methods can

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automatically learn the underlying map from observed images to HR-HSI. Existing deep learning methods based on fusion can be divided into two categories: one is supervised learning, the other is unsupervised learning. For a long time, supervised learning-based [12], [13], [14] convolutional neural networks have been demonstrated the state-of-the-art performance on HSI super-resolution task. For fully supervised learning, there is no doubt that large-scale training triplets (LR-HSI, HR-MSI, and HR-HSI) must be collected in advance for training the super-resolution network. There are some differences in the distribution of statistical characteristics in different datasets, and the network model obtained by fully supervised training lacks sufficient generalization ability in real application. To mitigate the dependence on HR-HSI as the supervised information and disadvantages of insufficient generalization, several works [15], [16], [17] have proposed the unsupervised deep learning paradigm. The works [15], [18] are devoted to designing two branch networks to extract feature information of LR-HSI and the corresponding HR-MSI. Qu et al. [15] design two autoencoder networks: one for extracting spectral bases from LR-HSI and the other for extracting spatial representations from HR-MSI. But in fact, these methods ignore the potential spectral mapping relationship between the observed images. To obtain more effective spatial-spectral information, Yao et al. [19] design a cross-attention module to extract and transfer significant spectral or spatial information. However, the high-frequency information extracted from a single spatial–spectral domain is limited. To explain the generalization behaviors of neural networks, many theoretical breakthroughs [20], [21], [22] have been made progressively by different model, data of frequency domain, and algorithm perspectives. This work [20] demonstrates convolutional neural networks tend to pay more attention to the structured information from phase components and keep robust to the variation of the amplitude. Given the amplitude and phase spectrum, they can then determine or reconstruct an image. In a general sense, the amplitude spectrum and phase spectrum represent the style and semantic information of the image, respectively. LR images are not sharp enough because of the lack of high-frequency details. Besides, as can be seen from Fig. 1, the phase spectrum and amplitude spectrum of image lacking high-frequency details are significantly different from those of HR images. Therefore, the frequency component difference between HR image and LR image can be reduced using amplitude spectrum and phase spectrum to reconstruct the image in frequency domain.

More high-frequency details are difficult to fully explore in a single spatial domain. The reconstruction of amplitude spectrum and phase spectrum from the perspective of frequency domain can minimize the spectral difference between HR-HSI and LR-HSI. Compared with the supervised method, the unsupervised method can make the model have excellent generalization ability, so as to adapt to the super-resolution reconstruction task of real scene.

With all the above considerations in mind, we propose the unsupervised across domain consistency-difference network (UACNet) for fusion-based HSI super-resolution, which aims at extracting the high-frequency details from observed images and effectively improve the generalization ability of unsupervised learning. To obtain rich multiscale feature information, we finely develop a triple-level encoder–decoder structure. We employ fast Fourier transformation to decompose latent features from triple-level encoder into amplitude features and phase features braches. The kernel of UACNet is a newly designed across domain consistency-difference (ADCD) module, which is dedicated to exploring dual domain high-frequency information. To flexibly adjust the receptive field and adapt to different domains, we embed the dilated causal convolution in the frequency domain processing branch. In summary, the main contributions of this work can be summarized as follows.

1) We present a new unsupervised network that reconstructs HR-HSI by the fusion of LR-HSI and HR-MSI. To the best of our knowledge, we are the first to adopt a network based on amplitude spectrum and phase spectrum tuning for HSI super-resolution task.
2) We develop the ADCD module to explore the across domain feature consistency-difference and dual-domain high-frequency details, which is preserved by memory units within the module. In addition, triple-level modeling allows the different scales feature aggregation, which provides abundant feature information for HR-HSI reconstruction.
3) Under the condition of dual-domain loss constraint, we conduct comprehensive experiments on public HSI datasets to analyze the proposed network. The experimental results demonstrate that the proposed model offers an improvement over the results of the state-of-the-art methods.

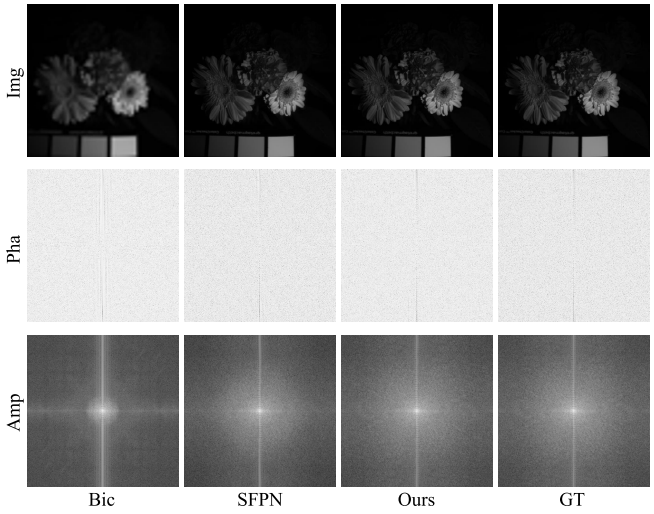


Fig. 1. (From left to right) They are the upsampled image after bicubic downsampling, the reconstructed image by SFPN method, the reconstructed image by the proposed method, and the original image. (From top to bottom) They are gray image, corresponding phase spectrum, and amplitude spectrum. This image is from the CAVE dataset, and the super-resolution scale is 8.

II. Related Work

A. Supervised Learning

In recent years, supervised learning has made significant progress in HSI super-resolution [12], [23], [24], [25]

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and natural image super-resolution [26], [27]. Combine the spatial and spectral details with an optimization problem, which is regarded as adaptively select an injection coefficient, Li et al. [28] employ a Laplacian pyramid network instead of the bicubic interpolation to upsample HSIs for the guided filtering-based MSI and HSI fusion. To effectively fuse the LR-HSI and HR-MSI, Han et al. [29] design a novel deep convolutional neural network architecture, which depends on the spectral and spatial correlation. Based on deep prior from the external and internal learning with spatial–spectral constraint, Zhang et al. [30] propose an effective network for coded HSI reconstruction. Pan et al. [31] develop a structure-color preserving network that employs the joint attention mechanism to describe and preserve structure and color information. Dong et al. [32] design the MoG-DCN, which is used to better exploit HSI-related domain knowledge and optimize deep convolutional neural network-based HSI super-resolution algorithm in a model-guided manner. To further optimize the model-guided network in an end-to-end manner, Wang et al. [33] propose the back-projection technique to extract the spatial and spectral details from the residual error to refine the reconstructed HR-HSI. Similarly, Li et al. [34] propose an enhanced back-projection method via spectral angle constraint, which is developed in fine stage to learn the content of spatial–spectral consistency, dramatically improving the performance gain. It is worth mentioning that transformer architecture [35], [36] and its various modifications have obtained outstanding achievements in many computer vision tasks. Because of the excellent ability of feature representations, especially by the self-attention in the transformer, Hu et al. [37] propose the Fusformer to globally explore the intrinsic relationship within features.

B. Unsupervised Learning

Although the supervised learning has been recognized to be effective for HSI super-resolution [38], unsupervised learning has gotten a lot of attention because it is closer to the process of human learning. Qu et al. [15] propose an unsupervised sparse Dirichlet network, which underexploits the information interactions between the images to be fused. To reduce the spectral distortion, the angular difference between representations is minimized by an unsupervised encoder–decoder architecture. Besides, they also propose an unregistered and unsupervised mutual dirichlet network [17] to exploit the uncharted problem domain of HSI super-resolution without the requirement of multimodality registration. Wang et al. [39] design an unsupervised fusion network, which models the joint full likelihood of the pixels in LR-HSI and HR-MSI parameterized by neural networks to describe the nonlinear mixture process from the latent abundances to the observed pixels. Wei et al. [40] propose an unsupervised recurrence-based method using pixel-aware refinement, which utilizes the intermediate reconstruction results to unsupervised learning. Yao et al. [19] propose a coupled unmixing network with a cross-attention mechanism, which is devised to yield more effective spatial–spectral information transfer in networks. Zhang et al. [41] develop a two-stage network composed of a fusion module and an adaptation module, to reconstruct the latent HR-HSI in a coarse-to-fine scheme. Similarly, Li et al. [34] propose dual-stage approach toward HSI super-resolution. In coarse stage, the coarse super-resolution image is obtained in band-by-band using supervised manner, while the coarse result is refined in a global perspective using unsupervised manner in fine stage. Based on the supervised coarse stage and the unsupervised fine stage, Li et al. [42] skillfully exploits the adjacent bands and RGB image to restore the current band without inputting all the bands.

C. Learning in Frequency Domain

At present, discrete cosine transform (DCT), wavelet transform (WT), and fast Fourier transform (FFT) are often used in frequency domain learning. This work [43] proposes a method of learning in the frequency domain and demonstrates its generality and superiority for a variety of tasks, including classification, detection, and segmentation. Shen et al. [44] propose a new mask representation by applying the DCT to encode the HR binary grid mask into a compact vector. Because of spatial redundancy within most filters in convolutional neural network, Liu et al. [45] propose a DCT-based frequency-domain dynamic pruning scheme to exploit the spatial correlations. Based on DCT, Qin et al. [46] rethink channel attention from the perspective of frequency analysis, prove that the conventional global average pooling is a special case of frequency domain feature decomposition, and propose a novel multifrequency channel attention. Frequency domain learning based on WT shows good performance in natural image super-resolution task [26], [47]. This article [47] presents a single-image super resolution method, which estimates wavelet detail coefficients of a desired LR image using a convolutional neural network on the given LR image. Compared to pixel domain, Jing et al. [48] prove that the frequency domain, especially high frequency, is more appropriate for image hiding by adopting WT, avoiding texture-copying artifacts and color distortion in pixel domain. Based on FFT, Chen et al. [20] explain the robustness performance of the convolutional neural network from the perspective of frequency domain.

III. Methodology

In this section, we present the proposed framework for HSI super-resolution, which is based on the fusion of LR-HSI and HR-MSI, and aims to explore dual domain high-frequency details to reconstruct HR-HSI. Our approach is shown in Fig. 2, which mainly contains three parts, triple-level encoder, ADCD network (ADCDN), and triple-level decoder.

A. Problem Formulation

Knowing the two images, one is an LR-HSI $Y \in \mathbb{R}^{h\times w\times B}$ and the other is HR-MSI $X \in \mathbb{R}^{H\times W\times b}$. The proposed super-resolution network aims to estimate a latent HR-HSI $Z \in \mathbb{R}^{H\times W\times B}$. $H(h)$, $W(w)$, and $B(b)$ represent the height, width, and spectral bands of the corresponding image, respectively. Generally speaking, $h \ll H$, $w \ll W$, and $b \ll B$.

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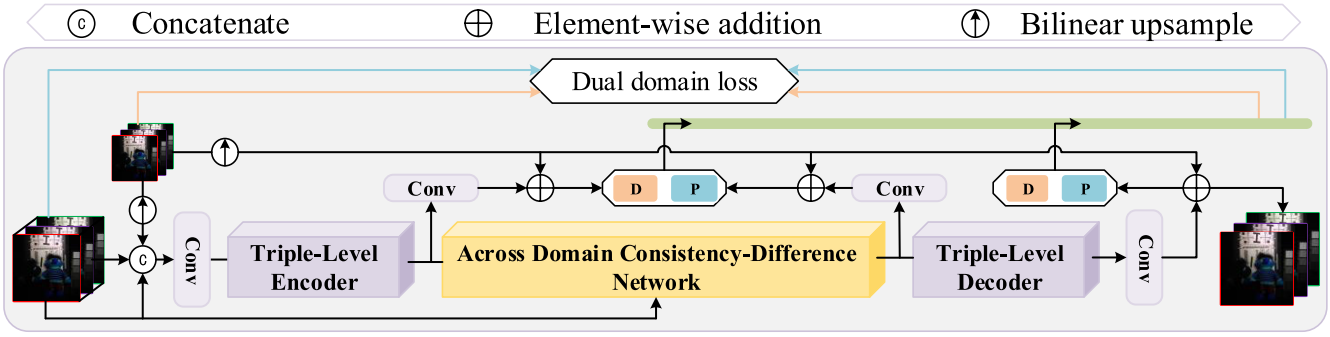


Fig. 2. Overview of the proposed UACNet. Based on triple-level encoder–decoder, we embed a cross-domain consistency-difference network dedicated to exploring and preserving high-frequency details in the dual domain. A spectral degradation function P and a spatial degradation function D are adopted to train the proposed network in an unsupervised manner. We calculate the loss of dual modality (LR-HSI, HR-MSI) in the pixel domain and the frequency domain.

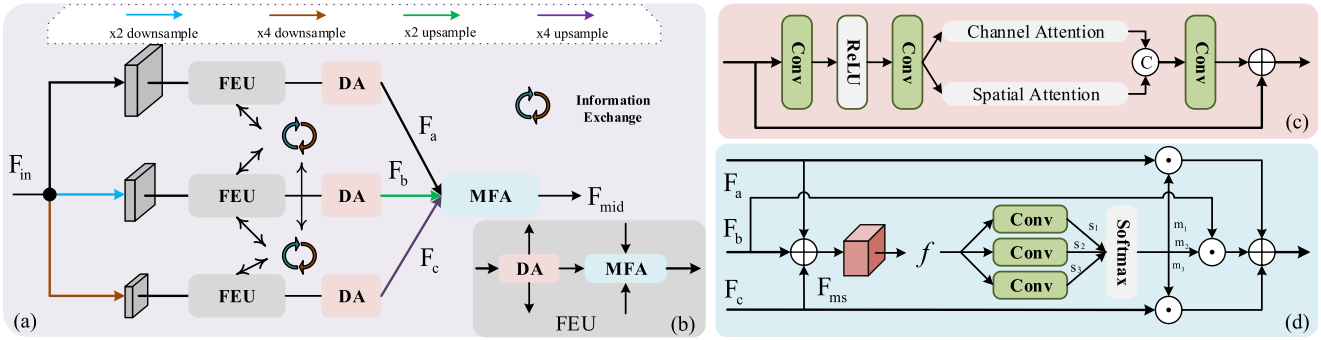


Fig. 3. Architecture of encoder. (a) Overview of the proposed triple-level encoder. (b) Detail of the FEU. Each FEU has three inputs. (c) Detail of the DA module. (d) Detail of the MFA module. "$\odot$" is the elementwise multiplication.

In addition, the relation between Z and the other two observations, X and Y, can be formulated as follows:

$$\begin{aligned} Y^{hw\times B} &= D^{hw\times HW} Z^{HW\times B} \\ X^{HW\times b} &= Z^{HW\times B} P^{B\times b} \end{aligned} \tag{1}$$

where $P \in \mathbb{R}^{B\times b}$ denotes the spectral degradation function from Z to X, and $D \in \mathbb{R}^{hw\times HW}$ is the spatial degradation function from Z to Y. When D and P are predefined, the loss function can be expressed as follows:

$$\mathcal{L} = \mathcal{L}_S\big((X, \hat{Z}P), (Y, D\hat{Z})\big) + \mathcal{L}_{\mathcal{F}}\big((X, \hat{Z}P), (Y, D\hat{Z})\big) \tag{2}$$

where $\hat{Z}$ is the reconstructed HR-his, $\mathcal{L}_S$ denotes the spatial domain loss, and $\mathcal{L}_{\mathcal{F}}$ represents the frequency domain loss.

The super-resolution process can be represented as follows:

$$\begin{aligned} F_{\text{in}} &= \text{Conv}(\text{cat}(X, \text{up}(Y), X)) \\ F_{\text{mid}} &= M_{\text{en}}(F_{\text{in}}) \\ F_{\text{mo}} &= M_{\text{net}}(F_{\text{mid}}, X) \\ F_{\text{out}} &= M_{\text{de}}(F_{\text{mo}}) \\ F_{\text{hr}} &= \text{Conv}(F_{\text{out}}) + \text{up}(Y) \end{aligned} \tag{3}$$

where cat denotes the "concatenate" operation, and up is the bilinear upsample. M_{en}, M_{net}, and M_{de} represents the encoder, the ADCDN, and the decoder, respectively.

B. Triple-Level Modeling

In this work, the encoder and decoder have the same architecture. Here, we select the encoder as an example to elaborate the triple-level modeling, which is composed of three parallel convolution streams with multiscale information exchange and shown in Fig. 3. As shown in Fig. 3(a), F_{in} is fed into three parallel branches, which are, respectively, maintained at full size, 2× downsampled, and 4× downsampled. Each branch consists of feature extraction unit (FEU) and dual attention (DA) modules. Each FEU includes DA and multiscale feature aggregation (MFA) module, the details of which are shown Fig. 3(b). Each FEU module has three inputs, one from the local branch and two from the output of DA in the FEU module of the remaining branches. Information exchange across parallel streams includes upsample and downsample operations to maintain spatial size matching, which consolidate the HR features with the help of LR features, and vice versa.

The detail of DA is shown in Fig. 3(c). The channel attention [49] and spatial attention [50] are widely used in low-level visual task, which suppresses less useful features and allows more informative features to pass further. The channel attention exploits the interchannel relationships by squeeze and excitation operations. For input feature X_{ca}, squeeze operation $F_{\text{sq}}(X_{\text{ca}})$ is carried out to compress the spatial dimension, while the number of feature channels remains unchanged. The excitation operation F_{ex} uses the W_{ca} parameter to generate weights for each feature channel. Then, the output A_c of the channel attention is obtained by

$$A_c = X_{\text{ca}} \cdot F_{\text{ex}}\big(F_{\text{sq}}(X_{\text{ca}}), W_{\text{ca}}\big). \tag{4}$$

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The spatial attention is designed to exploit the spatial dependencies, which aims to generate a spatial attention map and use it to rescale the input feature. For input feature X_{sa}, the spatial attention first applies global average pooling and max pooling along the channel dimensions and concatenates the outputs to obtain the feature map f_{sa}. Finally, f_{sa} is passed through a convolution and sigmoid activation function to obtain the spatial attention map, which is used to rescale the input feature X_{sa}. In this article, the channel attention and spatial attention have the same input.

The detail of MFA is shown in Fig. 3(d). The outputs of DA module in the three parallel streams are the input of MFA. Taking MFA at the end of the encoder as an example, we first combine these multiscale features using an elementwise sum as: $F_{\text{ms}} = F_a + F_b + F_c$. Then, the feature $f(F_{\text{ms}})$ passes three parallel convolution layers to obtain s_1, s_2, s_3. Here, the function f consists of three operations, first global averaging pooling, then 3×3 convolution, and finally PReLu activation function. The change process of feature size is as follows: $\mathbb{R}^{H\times W\times C} \to \mathbb{R}^{1\times 1\times C} \to \mathbb{R}^{1\times 1\times C/8}$. After three parallel convolution layers, s_1, s_2, s_3 have the same spatial size $\mathbb{R}^{1\times 1\times C}$. Next, the Softmax is used to s_1, s_2, s_3, and obtain the corresponding activation map m_1, m_2, and m_3. Finally, the overall process of MFA is defined as: $F_{\text{mid}} = m_1 F_a + m_2 F_b + m_3 F_c$.

In summary, both DA and MFA can be considered as attention mechanisms in a broad sense, focusing on important information in features. The DA module explores the pixel–pixel and channel–channel dependencies, while the MFA module further focuses on highly informative features from different branches and scales to improve the performance of the model.

C. Across Domain Consistency-Difference Network

The spatial domain is an image space composed of pixels, while the frequency domain uses frequency as the independent variable to describe the features of the image. In this article, we first interpolate the image to the same spatial dimensions as the HR image, and then process the pixels directly (in the spatial domain) and the amplitude spectrum and phase spectrum (in the frequency domain). The spatial domain to frequency domain transformation is realized by FFT. In this section, the ADCDN is examined in detail.

Given a latent feature tensor $x \in \mathbb{R}^{H\times W\times C}$, its Fourier transformation $\mathcal{F}$ is formulated as

$$\mathcal{F}(x)(m,n) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} x(h,w) e^{-j2\pi\left(\frac{h}{H}m + \frac{w}{W}n\right)} \tag{5}$$

where $\mathcal{F}(x)(m,n)$ denotes the value of the feature in the frequency domain corresponding to the coordinates m and n. The transformation can be implemented efficiently by applying the fast Fourier transformation algorithm. Naturally, we can obtain the corresponding amplitude spectrum $\mathcal{A}(x)$ and phase spectrum $\mathcal{P}(x)$

$$\begin{aligned}\mathcal{P}(x) &= \arctan\frac{I(\mathcal{F}(x)(m,n))}{R(\mathcal{F}(x)(m,n))}\\ \mathcal{A}(x) &= \sqrt{R(\mathcal{F}(x)(m,n))^2 + I(\mathcal{F}(x)(m,n))^2}\end{aligned} \tag{6}$$

where R and I denote the real and imaginary parts. In experiments, FFT is computed independently on each channel. In addition, the inverse Fourier transformation (IFT) $\mathcal{F}^{-1}(\mathcal{A}(x'), \mathcal{P}(x')) = x'$ can be used to reconstruct amplitude and phase spectrum features back into the spatial domain.

Algorithm 1 Framework of ADCDN (M_{net})

Input: High-resolution multispectral image X, the output of encoder F_{mid}
Output: F_{mo} (the input of decoder)

M_s denotes the signle ADCD module.
Set the number of M_s as t, $(t \geq 1)$.
Set $u_0 = X$.
Set $v_0 = F_{\text{mid}}$.
for $i = 0$ to $t-1$ **do**
  $u_{i+1} = \text{conv}(\text{conv}(u_i))$
  $v_{i+1} = M_s(v_i) + u_{i+1}$
Obtain the final output $F_{\text{mo}} = v_t$

Based on the reconstruction of amplitude and phase spectrum, we design the ADCDN, which is described in Algorithm 1. In a nutshell, the ADCD first learns the informative features from both frequency domain and spatial domain perspectives. Then, M_s adopts consistency learning and difference learning to enhance the consistency of dual-domain features and reduce intradomain difference.

The detail of single M_s is shown in Fig. 4. Because the amplitude component contains low-level information, while the high-level semantics is preserved in phase component, we utilize FFT to decompose features into amplitude spectrum and phase spectrum, and design two branch to adjust corresponding amplitude or phase information. The IFFT operation returns the reconstructed feature map from the frequency domain to the spatial domain. To flexibly adjust the receptive field and adapt to different domains, dilated causal convolution [51] is adopted to the above two branches. After refining amplitude spectrum and phase spectrum, we can obtain

$$\begin{aligned}\mathcal{A}_i &= \mathcal{A}(\text{FFT}(\text{Conv}(v_i)))\\ \mathcal{P}_i &= \mathcal{P}(\text{FFT}(\text{Conv}(v_i)))\\ F_A &= \text{IFFT}(f_s(\mathcal{A}_i), \mathcal{P}_i) + \text{Conv}(v_i)\\ F_P &= \text{IFFT}(\mathcal{A}_i, f_s(\mathcal{P}_i)) + \text{Conv}(v_i)\end{aligned} \tag{7}$$

where f_s denotes Conv-ReLu-Conv-Causal Conv block. Besides, we also deal with spatial domain features through convolution blocks $F_n = \text{Conv}(\text{ReLu}(\text{Conv}(v_i)))$. Then, we exploit the consistency-difference of feature from different branches. As shown in Fig. 4, C-L and D-L denote the consistency learning and difference learning, respectively. The consistency learning process can be formulated as follows:

$$\begin{aligned}F_C &= \text{Conv}(F_A \odot F_P \odot F_n)\\ F_{C1} &= \text{Conv}(F_C + F_A)\\ F_{C2} &= \text{Conv}(F_C + F_P)\\ F_C^{\text{AP}} &= \text{Conv}(F_{C1} + F_{C2})\end{aligned} \tag{8}$$

where $\odot$ is the elementwise multiplication. F_C^{AP} contains consistency boosting of features. Accordingly, the difference

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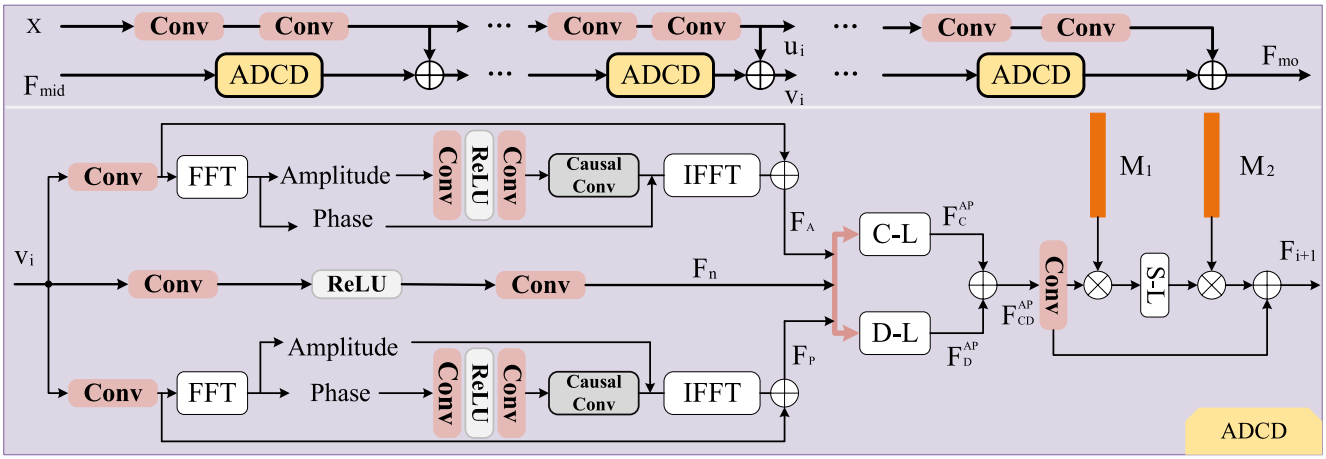


Fig. 4. Top part is the architecture of ADCDN. The bottom part is the details of single ADCD module. Both the cross-domain and cross-modality fusion feature are achieved and preserved by every single ADCD module. M_1, M_2 denotes the fully connected layers. "$\otimes$" denotes the matrix multiplication.

learning process can be formulated as follows:

$$\begin{aligned} F_D^{\mathrm{AP}} &= \mathrm{Conv}(|F_A - F_P| \odot F_n) \\ F_{\mathrm{CD}}^{\mathrm{AP}} &= \mathrm{Conv}\left(F_C^{\mathrm{AP}} + F_D^{\mathrm{AP}}\right) \end{aligned} \tag{9}$$

where $|\cdot|$ calculates the absolute value, F_D^{AP} denotes the difference of cross domain feature, and $F_{\mathrm{CD}}^{\mathrm{AP}}$ achieves richer feature information. Finally, we adopt two independent of input and learnable fully connected layers as memory units, which is used to store and provide long-range dependence information. The process can be expressed as follows:

$$F_{i+1} = M_2\left(F_{\text{S-L}}\left(M_1\left(\mathrm{Conv}\left(F_{\mathrm{CD}}^{\mathrm{AP}}\right)\right)\right)\right) + \mathrm{Conv}\left(F_{\mathrm{CD}}^{\mathrm{AP}}\right) \tag{10}$$

where $F_{\text{S-L}}$ denotes the Softmax and $L1$ normalization and $v_{i+1} = F_{i+1} + u_{i+1}$ is the output of single ADCD module (M_s). Naturally, we get the reconstructed HR-HSI by sending the output (F_{mo}) of M_{net} to the decoder, which is the same as encoder.

D. Dual-Domain Loss Function

We train the proposed network in an unsupervised manner. Therefore, we adopt the spatial degradation D and spectral degradation P. Let $\hat{Z}$ denotes the reconstructed HR-HSI. The spatial domain loss is defined as follows:

$$\mathcal{L}_S = \left\|X - \hat{Z}P\right\|_1 + \left\|Y - D\hat{Z}\right\|_1. \tag{11}$$

Meanwhile, the frequency domain loss is defined as follows:

$$\begin{aligned} \mathcal{L}_{\mathcal{A}} &= \left\|\mathcal{A}(X) - \mathcal{A}\left(\hat{Z}P\right)\right\|_1 + \left\|\mathcal{A}(Y) - \mathcal{A}\left(D\hat{Z}\right)\right\|_1 \\ \mathcal{L}_{\mathcal{P}} &= \left\|\mathcal{P}(X) - \mathcal{P}\left(\hat{Z}P\right)\right\|_1 + \left\|\mathcal{P}(Y) - \mathcal{P}\left(D\hat{Z}\right)\right\|_1 \\ \mathcal{L}_{\mathrm{fft}} &= \left\|\mathcal{F}(X) - \mathcal{F}\left(\hat{Z}P\right)\right\|_1 + \left\|\mathcal{F}(Y) - \mathcal{F}\left(D\hat{Z}\right)\right\|_1 \\ \mathcal{L}_{\mathcal{F}} &= \lambda_{\mathcal{A}}\mathcal{L}_{\mathcal{A}} + \lambda_{\mathcal{P}}\mathcal{L}_{\mathcal{P}} + \lambda_f\mathcal{L}_{\mathrm{fft}} \end{aligned} \tag{12}$$

where $\lambda_{\mathcal{A}}$, $\lambda_{\mathcal{P}}$, and λ_f are empirically set to 0.01, 0.01, and 0.05, respectively. The dual domain loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_S + \mathcal{L}_{\mathcal{F}}. \tag{13}$$

In training process, we minimize the dual-domain loss in three stages

$$\begin{aligned} \hat{Z}_1 &= \mathrm{Conv}(M_{\mathrm{en}}(F_{\mathrm{in}})) + \mathrm{up}(Y) \\ \hat{Z}_2 &= \mathrm{Conv}(M_{\mathrm{net}}(F_{\mathrm{mid}})) + \mathrm{up}(Y) \\ \hat{Z}_3 &= \mathrm{Conv}(M_{\mathrm{de}}(F_{\mathrm{mo}})) + \mathrm{up}(Y) \end{aligned} \tag{14}$$

where $\hat{Z}_1$, $\hat{Z}_2$, and $\hat{Z}_3$ are reconstructed HR-HSIs in the three stages, respectively. In this work, we select $\hat{Z}_3$ as the final result. Thus, the total loss $\mathcal{L}_{\mathrm{total}}$ used to train the proposed network is expressed as follows:

$$\begin{aligned} \mathcal{L}_1 &= \mathcal{L}\left(\left(X, \hat{Z}_1P\right), \left(Y, D\hat{Z}_1\right)\right) \\ \mathcal{L}_2 &= \mathcal{L}\left(\left(X, \hat{Z}_2P\right), \left(Y, D\hat{Z}_2\right)\right) \\ \mathcal{L}_3 &= \mathcal{L}\left(\left(X, \hat{Z}_3P\right), \left(Y, D\hat{Z}_3\right)\right) \\ \mathcal{L}_{\mathrm{total}} &= \lambda_1\mathcal{L}_1 + \lambda_2\mathcal{L}_2 + \lambda_2\mathcal{L}_3 \end{aligned} \tag{15}$$

where $\lambda_1, \lambda_2, and \lambda_3$ are set to 0.1, 0.1, and 0.8, respectively.

IV. Experiment

A. Datasets

Following [41], we use three public HIS datasets, CAVE [52], Harvard [53], and ICVL dataset [54] to evaluate the proposed method. CAVE dataset includes 32 HSIs, each of which is of size 512 × 512. Each image consists 31 spectral bands in a wavelength range of 400–700 nm at 10-nm step. We randomly select 12 HSIs for training and the rest 20 HSIs for test the performance of the proposed model. The Harvard dataset contains 50 HSIs from outdoor and indoor in the spectral range is from 420 to 720 nm at 10-nm step. The resolution of each image is 1392 × 1040. About 20 HSIs are selected as training set, 30 HSIs are used for testing. In the ICVL dataset, we randomly choose 70 HSIs each of which is of size 1392 × 1300 × 31. One image has 31 bands in the spectral range from 400 to 700 nm. In addition, 20 HSIs are randomly selected as a training set, and the remaining 50 HSIs as test set. The Chikusei dataset [55] has 128 spectral bands in the spectral range from 363 to 1018 nm and 2517 × 2335 pixels in total. The Pavia University dataset [1] consists

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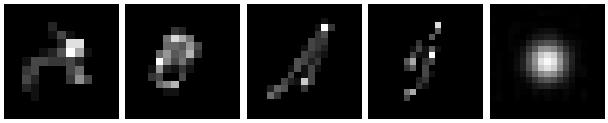


Fig. 5. Five blur kernels utilized to produce the degradation D in test phase.

of 610 × 340 pixels with a ground sampling distance of 1.3 m. The 12 noisy bands have been removed so a total of 103 bands. We select the top-left area of Chikusei dataset and Pavia University dataset with a spatial size of 448 × 512 and 320 × 224 for testing.

B. Comparison Methods and Evaluation Metrics

To demonstrate the superiority of proposed method, six method for HSI super-resolution are employed, including NSSR [56], MHFNet [57], PZRes-Net [58], UAL [41], WANG [59], and SFPN [42]. NSSR [56] exploits the sparse or manifold prior of the latent HSI. UAL [41] and SFPN [42] belong to two-stage training methods, the first stage is supervised manner, the second stage is unsupervised, while MHFNet [57], PZRes-Net [58], and WANG [59] are single-stage networks. For UAL, we only use the model proposed in its paper for comparative experiment, and other details are consistent with our method. Unless otherwise stated, the SFPN method is compared with our unsupervised method using a coarse model trained in a supervised manner. For a comprehensive evaluation, we adopted four commonly used quantitative metrics: root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), and structural similarity index (SSIM).

TABLE I
PERFORMANCE OF DIFFERENT BLUR KERNELS ON CAVE DATASET WITH SCALE ×8

Methods	kernel	RMSE↓	PSNR↑	SAM↓	SSIM↑
NSSR [56]	k1	8.62	30.29	11.97	0.8602
	k2	9.07	29.87	12.39	0.8530
	k3	8.60	30.33	11.76	0.8628
	k4	8.79	30.14	12.03	0.8592
	k5	8.49	30.42	11.83	0.8625
MHFNet [57]	k1	4.16	36.78	8.74	0.9555
	k2	4.43	36.29	8.98	0.9519
	k3	4.07	37.00	8.68	0.9563
	k4	4.23	36.69	8.82	0.9546
	k5	4.16	36.73	8.73	0.9551
PZRes-Net [58]	k1	3.01	39.89	6.15	0.9771
	k2	3.07	39.47	6.38	0.9751
	k3	2.86	40.06	6.15	0.9744
	k4	2.96	39.80	6.25	0.9765
	k5	2.91	39.94	6.13	0.9738
UAL [41]	k1	2.49	41.26	4.81	0.9853
	k2	2.56	41.03	4.91	0.9846
	k3	2.46	41.34	4.77	0.9855
	k4	2.51	41.20	4.84	0.9852
	k5	2.47	41.33	4.75	0.9855
WANG [59]	k1	2.30	41.79	4.98	0.9841
	k2	2.39	41.52	5.07	0.9832
	k3	2.26	41.97	4.92	0.9845
	k4	2.31	41.77	5.01	0.9839
	k5	2.25	41.98	4.85	0.9847
SFPN [42]	k1	2.22	42.05	4.46	0.9852
	k2	2.24	42.02	4.48	0.9849
	k3	2.16	42.15	4.43	0.9854
	k4	2.53	40.85	4.73	0.9837
	k5	2.32	41.98	4.31	0.9849
Ours	k1	**1.88**	**43.50**	**3.82**	**0.9893**
	k2	**1.95**	**43.22**	**3.91**	**0.9886**
	k3	**1.79**	**43.88**	**3.64**	**0.9901**
	k4	**1.86**	**43.55**	**3.81**	**0.9893**
	k5	**1.81**	**43.78**	**3.68**	**0.9898**

C. Implementation Details

For synthetic datasets CAVE, Harvard and ICVL, we use the spectral response function of a Nikon D700 camera as spectral degradation P and take five Gaussian kernel as spatial degradation D in training, respectively. Landsat-8 is selected for Chikusei and Pavia University dataset as the spectral response function.

To simulate various degenerations in practice and demonstrate the generalization ability of the proposed model, we employ four non-Gaussian kernels and one Gaussian kernel for testing, as shown in Fig. 5. After determining the spatial degradation and spectral degradation, we can obtain the corresponding LR-HSI. Given a training HR-HSI, we extract patches of size 128 × 128 with stride 64 to generate training pairs. The super-resolution scales are set as 4, 8, and 16. In this work, we implement the UACNet in Pytorch with Titan Xp graphics processing unit (GPU). To train the proposed model, we adopt the Adam optimizer. The initial learning rate is set to 1×10^{-4} and decayed to 0.8 times its original value every ten epochs.

D. Ablation Study

1) Generalizability to Various Blur Kernels: To analyze the generalization ability of the proposed model under various degradation conditions, experiments are performed on CAVE dataset. At the same time, the super-resolution scale factor is set to 8. When training the proposed model, Gaussian blur kernels with sizes 7 × 7, 8 × 8, 9 × 9, 13 × 13, and 15 × 15 are randomly adopted. During the test, we randomly selected the blur kernel as shown in Fig. 5, whose sizes are 13 × 13, 15 × 15, 17 × 17, 19 × 19, and 19 × 19, respectively. Table I shows the numerical results of different methods under five different blur kernels. We compare the proposed model with the methods mentioned above, and the best values are marked in bold. Obviously, our proposed approach outperforms better than the other methods on all indicators. In detail, the PSNR value of the proposed method is 1.73 dB higher than the second best model under k3 blur kernel degradation. Besides, the reconstructed HR-HSIs and the corresponding error maps of one test image in CAVE dataset with scale ×8 are shown in Fig. 6. We select the spectral bands 9-19-29 as RGB to show the visual performance. The significant superiority of the proposed method over the state-of-the art methods is further validated. Our method consistently produces the smallest errors over all the image, and there are only subtle errors, which are nearly invisible. There are obvious errors between the HR image reconstructed by NSSR method and the ground truth. The model performance of SFPN is second only to that of our proposed method.

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TABLE II
PERFORMANCE ON HARVARD AND ICVL DATASET FOR SCALE ×8

	Harvard							ICVL						
	NSSR	MHFNet	PZRes-Net	UAL	WANG	SFPN	Ours	NSSR	MHFNet	PZRes-Net	UAL	WANG	SFPN	Ours
RMSE ↓	2.72	1.90	1.81	1.75	1.69	1.63	**1.61**	2.14	1.29	1.14	1.06	0.93	0.85	**0.78**
PSNR ↑	39.84	42.34	43.51	43.84	44.34	44.62	**44.80**	41.82	44.61	46.35	47.03	47.49	49.17	**49.59**
SAM ↓	3.54	2.84	2.78	2.63	2.50	**2.43**	2.49	1.29	0.96	0.91	0.85	0.81	0.78	**0.73**
SSIM ↑	0.9703	0.9812	0.9824	0.9837	0.9841	0.9742	**0.9844**	0.9803	0.9953	0.9954	0.9958	0.9961	0.9968	**0.9971**

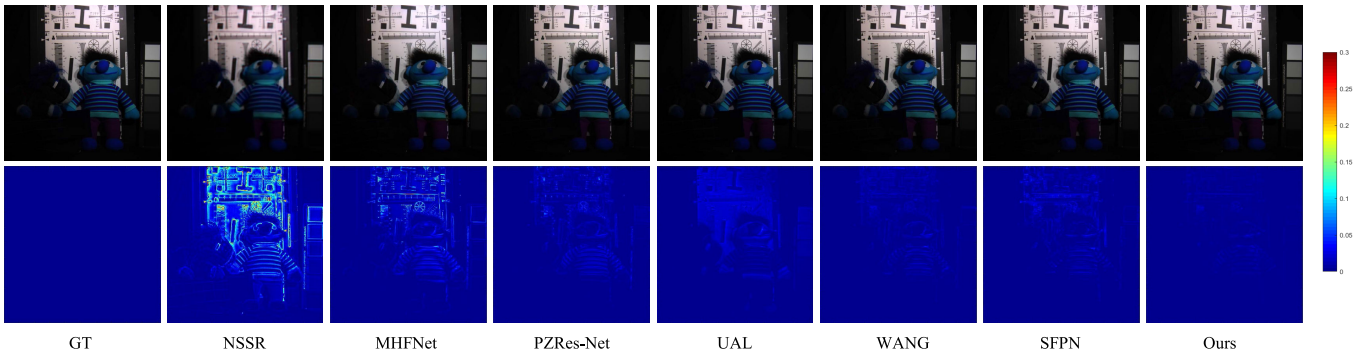


Fig. 6. Visual results and reconstruction error maps of an image in CAVE dataset with scale ×8. The first row is the reconstructed results with spectral bands 9-19-29 as RGB. The second is the corresponding error maps.

TABLE III
PERFORMANCE ON VARIOUS DATASET FOR SCALE ×4

	CAVE				Harvard				ICVL			
	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
NSSR	5.17	34.89	7.07	0.9090	2.28	41.61	3.14	0.9736	1.93	43.21	1.135	0.9834
MHFNet	3.30	38.91	5.82	0.9709	1.74	44.01	2.67	0.9821	0.95	47.87	0.95	0.9880
PZRes-Net	2.41	39.13	5.64	0.9756	1.72	44.16	2.59	0.9835	0.80	48.24	0.82	0.9923
UAL	2.00	43.12	4.07	0.9887	1.68	44.45	2.46	0.9852	0.84	49.05	0.75	0.9961
WANG	1.62	44.70	3.67	0.9905	1.59	44.86	2.41	0.9853	0.79	49.53	0.71	0.9970
SFPN	1.55	45.17	3.37	0.9913	1.53	45.11	2.37	0.9855	0.75	49.84	0.70	0.9974
Ours	**1.42**	**45.79**	**3.16**	**0.9926**	**1.52**	**45.19**	**2.36**	**0.9859**	**0.71**	**50.14**	**0.69**	**0.9974**

2) Generalizability to Different Datasets, Scales, and Models: In this section, the proposed model is compared with mentioned methods on three datasets to verify the generalizability of the model. In the experiments, the blur kernel k3 is employed to test the model performance. In subsequent experiments, unless otherwise specified, the blur kernel used in the test phase is k3. The average RMSE, PSNR, SAM, and SSIM results on the Harvard and ICVL dataset are shown in Table II for scale 8. We can see that the proposed method outperforms other deep learning-based methods by a large margin. The Harvard dataset is more challenging than the CAVE dataset due to the higher resolution and more complex scenario. The average PSNR gain over the SFPN method (the second best) is 0.18 dB for Harvard dataset. For ICVL dataset, the proposed method outperforms the SFPN method (the second best) by 0.42 dB on average.

Table III shows the average results on the three datasets by the competing methods for scale factor 4. As can be observed, the proposed method reconstructs more accurate results, such as the proposed method can avoid spectral distortion well, which has a small SAM value for used HSI datasets. For example, in the CAVE dataset of ×4 super-resolution reconstruction, SAM is 7.07 for the NSSR method and 3.37 for the SFPN. Our proposed method is 3.16, which is 0.21 lower than the second best method. In addition, the experimental results on the three dataset of different methods for scale ×16 are listed in Table IV. Although these models based on the fusion approach have good performance, compared with our approach, they are slightly inferior. Taking the ICVL dataset as an example, the PSNR value of our method is 0.17 dB higher than the second-best method. Selecting the spectral bands 9-19-29 as RGB, Fig. 7 shows the error maps of the ×8 reconstructed HSI from Harvard dataset. For the two test images, the NSSR method performs the worst, UAL has a slightly better performance than PZRes-Net, and the performance of our method obviously outperforms the other methods. Besides, Fig. 8 shows the super-resolved ×8 error map results for one image from ICVL dataset. To more comprehensively verify the superiority of the proposed method, we show the PSNR of single spectral band of HR-HSI reconstructed by different methods in Fig. 9. Similarly, Fig. 10 exhibits the super-resolved error map results for another image from ICVL dataset. We find that some images appear smoothing or ringing results, which have a large margin

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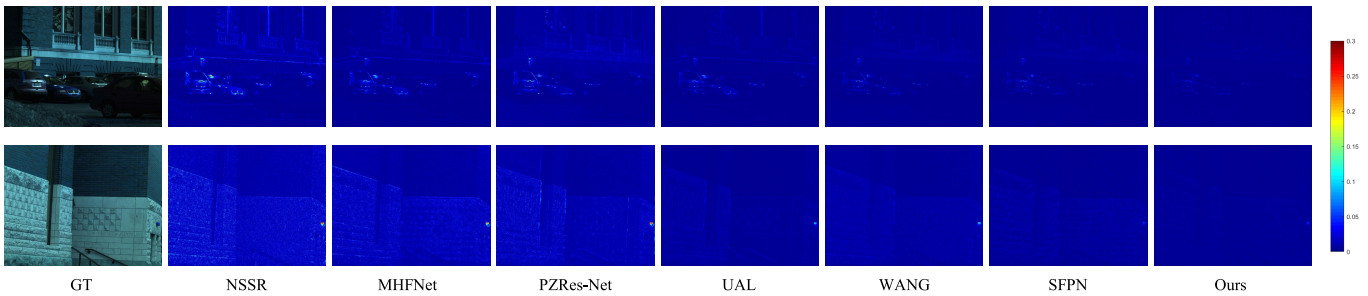


Fig. 7. Error maps of reconstructed images by different methods. The two images from Harvard dataset and the super-resolution scale is set to ×8. The first column images are the ground truth with spectral bands 9-19-29 as RGB.

TABLE IV
PERFORMANCE ON VARIOUS DATASET FOR SCALE ×16

	CAVE				Harvard				ICVL			
	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
NSSR	9.46	29.55	14.3	0.8646	2.91	38.81	3.84	0.9655	2.57	40.17	1.41	0.9794
MHFNet	4.95	35.37	9.34	0.9439	2.43	41.20	3.53	0.9761	2.08	42.62	1.14	0.9963
PZRes-Net	3.42	38.14	6.73	0.9504	2.36	41.42	3.46	0.9780	1.46	45.58	0.99	0.9967
UAL	3.13	39.45	6.48	0.9653	2.11	43.11	2.81	0.9814	1.25	46.11	0.95	0.9964
WANG	2.93	39.87	6.14	0.9769	1.92	43.57	2.74	0.9835	1.18	46.64	0.92	0.9959
SFPN	2.51	40.84	5.56	0.9814	1.85	43.79	2.73	0.9836	1.05	47.76	0.89	0.9966
Ours	**2.44**	**41.44**	**5.13**	**0.9836**	**1.81**	**43.94**	**2.73**	**0.9837**	**1.00**	**47.93**	**0.86**	**0.9970**

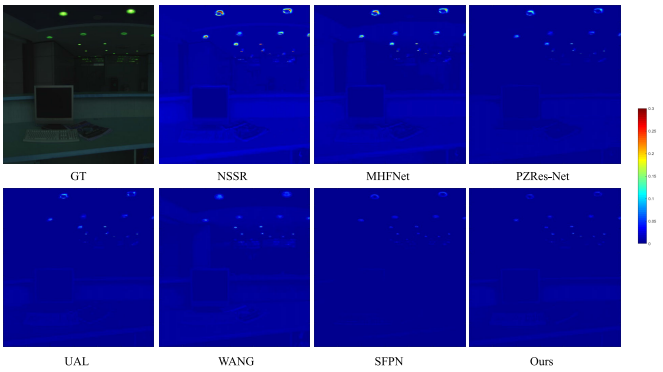


Fig. 8. Reconstruction error maps of an image from ICVL dataset with scale ×8. The first image is the ground truth with spectral bands 9-19-29 as RGB. The others are the corresponding error maps.

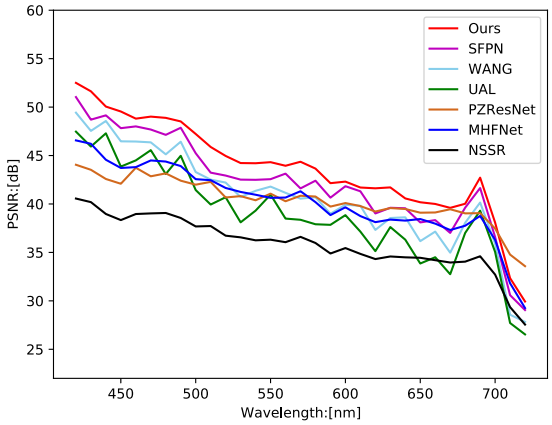


Fig. 9. Quantitative comparisons of the proposed model with the state-of-the-art methods in terms of the PSNR value of each spectral band of the reconstructed HR-HSI. This image "imgb4" is selected from Harvard dataset.

TABLE V
PERFORMANCE ON CHIKUSEI DATASET FOR SCALE ×8

	NSSR	MHFNet	PZRes-Net	UAL	WANG	SFPN	Ours
RMSE ↓	3.80	2.48	1.94	1.58	1.56	1.53	1.51
PSNR ↑	36.49	40.12	42.18	43.82	44.01	44.17	44.28
SAM ↓	11.05	9.27	9.35	9.19	9.11	9.07	9.04
SSIM ↑	0.9412	0.9656	0.9857	0.9900	0.9914	0.9905	0.9907

of error. However, the proposed model recovers high-quality results.

Moreover, we verify the performance of the proposed method on remote sensing datasets, Chikusei dataset, and Pavia University. Table V and Fig. 11, respectively, show the numerical results and visual representation of different methods on Chikusei test data. The superior performance of the proposed method on SAM index is not obvious, but it is advanced on PSNR index. The semantic information of HSI plays an important role in downstream tasks. Therefore, we show the ×4 super-resolution amplitude spectrum of Pavia University test data and the gray image of the pseudo-color image corresponding to the 61-21-1 spectral band in Fig. 12. We can find that the HR-HSI reconstructed by the proposed model is almost the same as that of ground truth.

To examine the applicability and demonstrate the generalization ability of the proposed UACNet on the CAVE dataset, we obtain three versions of UACNet, i.e., "UACNet(CAVE)," "UACNet(Harvard)," and "UACNet(ICVL)," in which the model is trained on the training set of the CAVE, Harvard, and ICVL datasets, respectively. The numerical results are shown in Table VI. It can be seen that the test performance of the models trained by different datasets does not show a huge drop on CAVE testing data for UACNet.

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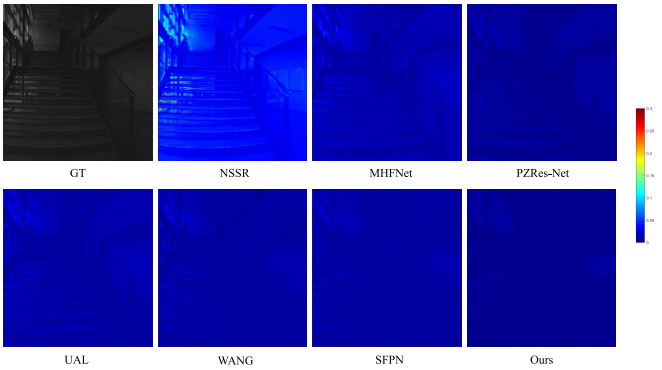


Fig. 10. Reconstruction error maps of an image from ICVL dataset with scale ×8. The first image is the ground truth with spectral bands 9-19-29 as RGB. The others are the corresponding error maps.

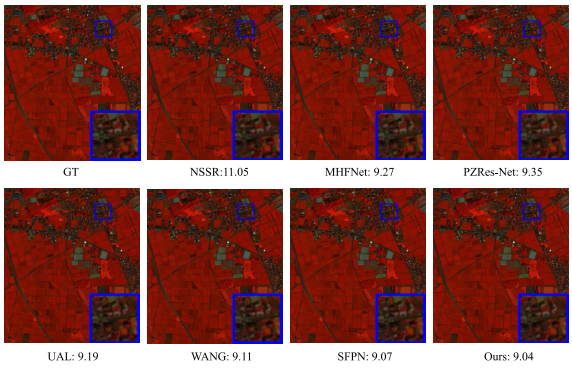


Fig. 11. Visual results of test data from Chikusei dataset with scale ×8 and spectral bands 62-40-10 as RGB. The numbers in the figure represent SAM values.

TABLE VI
ABLATION STUDIES OF THE MODEL GENERALIZATION ABILITY

	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
UACNet (CAVE)	1.79	43.88	3.64	0.9901
UACNet (Harvard)	2.49	41.01	5.46	0.9803
UACNet (ICVL)	2.59	40.81	4.84	0.9815

TABLE VII
ABLATION STUDIES OF THE NUMBER OF M_s

Number of M_s	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
0	1.84	43.63	3.74	0.9894
1	1.82	43.78	3.69	0.9899
3	1.79	43.88	3.64	0.9901
4	1.78	43.93	3.65	0.9901

3) Study of M_s: The ADCDN M_{net} consists of multiple ADCD modules M_s and multiple convolution layers for feature extraction of MSIs. To verify the validity of the module M_s and analyze how many highly similar modules M_s are selected to facilitate the learning of context content, this section sets different numbers of M_s for the whole ADCDN M_{net}. We perform the experiments on the CAVE dataset, and evaluate the super-resolution performance with scale ×8. Table VII depicts the effect of different numbers of M_s on the performance of the model. It can be observed from Table VII that when the number of M_s involved in the establishment increases, the values of four metrics are significantly improved. It indicates that the more the number of participating M_s, the more conducive to model learning. In detail, when the proposed model does not use the ADCDN, that is $M_s = 0$, the average PSNR of the reconstructed images is 43.63 dB. PSNR increased by 0.15 dB when only one of the modules was used. However, there is no doubt that too many M_s cause parameters to be added during exploiting the dual-domain high-frequency details. Under the compromise consideration, three M_s are picked to complete the experiments in this article.

TABLE VIII
ABLATION STUDIES OF THE NUMBER OF CHANNELS

Number of C	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
64	1.45	45.66	3.16	0.9925
96	1.42	45.79	3.16	0.9926
128	1.37	46.08	3.05	0.9931

TABLE IX
MODEL PARAMETER COMPARISON

Methods	Pram	Methods	Pram
NSSR	-	MHFNet	1.7M
PZRes-Net	0.7M	UAL	7.1M
SFPN	1.2M	Ours	14.2M

TABLE X
ABLATION STUDIES OF THE DUAL-DOMAIN LOSS

supervision	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
$\mathcal{L}_S$	1.49	45.39	3.28	0.9920
$\mathcal{L}_\mathcal{F}$	1.47	45.55	3.33	0.9919
$\mathcal{L}_S + \mathcal{L}_\mathcal{F}$	1.42	45.79	3.16	0.9926

4) Study of Channels Size: To analyze the effect of channel size on the performance of the proposed method, we conducted experiments on CAVE dataset with ×4 super-resolution. The channel size represents the channel size of encoder–decoder (M_{en}-M_{de}) and M_{net} after passing through the convolutional layer. In the experiment, we set the number of model channels as 64, 96, and 128, respectively. Table VIII reports the results of different number of channels, validating the potential of the proposed model. It can be seen that as we widen the model, the model performance increases within a certain range. However, considering the model parameters and calculation costs, except for this experiment, the channel size is set at 96. In addition, Table IX shows the number of parameters for the different models. PZRes-Net model is the smallest, of course, the model performance is limited to a certain extent. The number of parameters in UAL model is half of that in UACNet model.

5) Study of Dual-Domain Loss: As aforementioned, we used a dual-domain loss to constrain the model training process by simultaneously exploring the latent high-frequency details from spatial domain and frequency domain. We verify the effect of dual-domain loss by conducting experiments on CAVE dataset with ×4 super-resolution. The proposed models are trained to use only spatial loss $\mathcal{L}_\mathcal{S}$, frequency domain loss $\mathcal{L}_\mathcal{F}$ and dual domain loss $\mathcal{L}_\mathcal{F} + \mathcal{L}_\mathcal{F}$, respectively. By comparing the first and the third rows of Table X, the effectiveness of dual loss can be validated. In detail, compared

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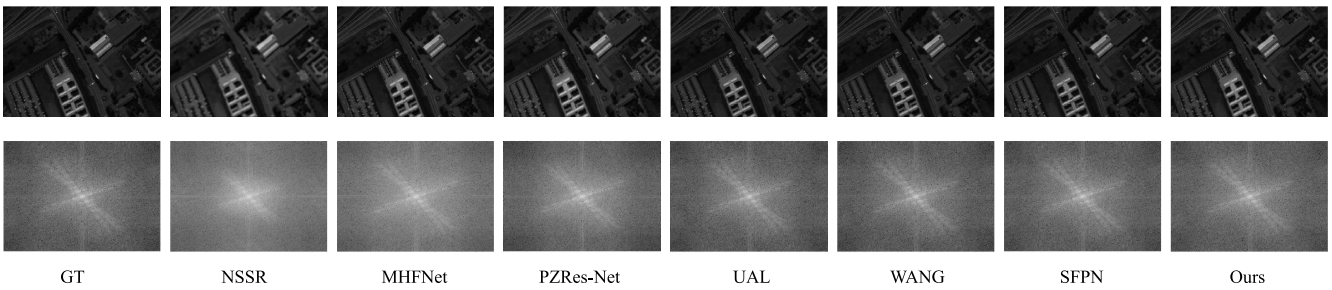


Fig. 12. Gray image of pseudo color image (61-21-1 spectral bands) and the corresponding amplitude spectrum. The test data are from the Pavia University dataset and the super-resolution scale is 4.

TABLE XI
ABLATION STUDIES OF BALANCE FACTOR

	λ_1:λ_2:λ_3=0.1:0.1:0.8		
	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.05
PSNR ↑	31.91	31.91	31.92
SAM ↓	4.16	4.14	4.09
	λ_1:λ_2:λ_3=0.2:0.2:0.6		
	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.05
PSNR ↑	31.87	31.86	31.78
SAM ↓	4.17	4.16	4.16
	λ_1:λ_2:λ_3=0.3:0.3:0.4		
	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.1	$\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f 0.1:0.1:0.05
PSNR ↑	31.79	31.77	31.79
SAM ↓	4.275	4.23	4.24

TABLE XII
ABLATION STUDIES ON CAVE DATASET FOR ×8

	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
SFPN(CoarseNet)	2.16	42.15	4.43	0.9854
SFPN(CoarseNet+FineNet)	1.95	43.50	4.24	0.9863
UACNet	1.79	43.88	3.64	0.9901
EACNet	1.54	45.29	3.34	0.9925

with the dual-domain loss function, the PSNR decreased by 0.4 dB using only spatial loss $\mathcal{L}_{\mathcal{S}}$. Using only the frequency domain loss $\mathcal{L}_{\mathcal{F}}$, PSNR decreased by 0.24 dNB. In addition, the proposed model can effectively avoid spectral distortion under the supervision of dual-domain loss, which is shown by SAM value. To verify the effect of balance factors in $\mathcal{L}_{\mathcal{S}}$, $\mathcal{L}_{\mathcal{F}}$, and $\mathcal{L}_{\text{total}}$, we conduct experiments on Pavia University dataset with ×4 super-resolution. Set λ_1:λ_2:λ_3 as 0.1:0.1:0.8, 0.2:0.2:0.6, 0.3:0.3:0.4 and $\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f as 0.1:0.1:1, 0.1:0.1:0.1, 0.1:0.1:0.05, respectively, for cross experimental verification. The experimental results of different balance factors settings are shown in Table XI. Obviously, when λ_1:λ_2:λ_3 = 0.1:0.1:0.8 and $\lambda_{\mathcal{A}}$:$\lambda_{\mathcal{P}}$:λ_f = 0.1:0.1:0.05, the model gets the best performance.

6) Discussion: It is particularly noted that the SFPN method is trained in a supervised manner in the above experiments, that is, the coarse model. To demonstrate the superiority of the proposed method, we further performed and presented convincing experiments on the CAVE dataset ×8 super-resolution task. As shown in Table XII, when comparing the SFPN coarse model (supervised training) with the proposed method (unsupervised training), the proposed method has better PSNR value than SFPN coarse model. The proposed unsupervised method is superior to supervised method SFPN coarse model for the following three reasons. First, the joint spatial-frequency domain reconstruction can obtain more abundant high-frequency details. Second, different from unsupervised single HSI super-resolution, the proposed method is based on MSI-HSI fusion super-resolution method. As one of the constraints of unsupervised methods, MSI narrow the gap between unsupervised and supervised methods in the traditional sense. Third, the proposed model has larger parameters than the supervised method. The details can be seen in Table IX. In addition, to compare and analyze the model performance comprehensively and fairly, we propose an enhanced model. The enhanced cross-domain consistency-difference network (EACNet) does not change any network structure and only uses supervised manner for training. Table XII shows that both EACNet and UACNet are superior to SFPN (CoarseNet + FineNet).

V. CONCLUSION

This article presents a novel unsupervised deep network for HSI super-resolution, which explores latent high-frequency details from the dual-domain perspective. In detail, the encoder–decoder adopts triple-level modeling to obtain abundant feature information. Meanwhile, an ADCDN is designed to adaptively adjust phase spectrum with semantic feature and amplitude spectrum with style feature, so as to achieve high super-resolution performance. The experimental results show that the proposed method is superior to the existing methods in quantitative and qualitative evaluation. The parameter number of the proposed model is about 14 M, so it is limited and inconvenient to deploy the model on mobile devices and practical applications. This is the main disadvantage of the proposed method. On the premise of maintaining the performance of the proposed method, the main direction of future work is to reduce the number of parameters in the model, such as by quantization.

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Zhiling Guo received the B.Sc. degree in electronic and information engineering from the Tianjin University of Technology, Tianjin, China, in 2021. He is currently pursuing the M.S. degree with the School of Telecommunications Engineering, Xidian University, Xi'an, China.

His research interests include computer vision and machine learning.



Jingwei Xin received the B.Sc. degree in electronic information science and technology from Shaanxi Normal University, Xi'an, China, in 2016, and the Ph.D. degree in information and telecommunications engineering from Xidian University, Xi'an, in 2021.

He is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. He has broad research interests in machine learning, pattern recognition, and computer vision. His research has been published in the academic venues and prestigious journals, such as European Conference on Computer Vision (ECCV) and *International Journal of Computer Vision* (IJCV).



Nannan Wang (Member, IEEE) received the B.Sc. degree in information and computation science from the Xi'an University of Posts and Telecommunications, Xi'an, China, in 2009, and the Ph.D. degree in information and telecommunications engineering from Xidian University, Xi'an, in 2015.

He is currently a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. He has authored over 150 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), *International Journal of Computer Vision* (IJCV), Conference on Computer Vision and Pattern Recognition (CVPR), and International Conference on Computer Vision (ICCV). His research interests include computer vision and machine learning.



Jie Li received the B.Eng. degree in electronic engineering, the M.Sc. degree in signal and information processing, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 1995, 1998, and 2004, respectively.

She is currently a Professor with the School of Electronic Engineering, Xidian University. Her research interests include image processing and machine learning. In these areas, she has authored approximately 50 technical articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), and *Information Sciences*.



Xiaoyu Wang (Fellow, IEEE) received the B.E. degree from the University of Science and Technology of China, Hefei, China, in 2006, and the M.S. degree in statistics and the Ph.D. degree in electrical and computer engineering from the University of Missouri, Columbia, MO, USA, in 2012.

From 2012 to 2015, he was a Research Scientist at NEC Laboratories America, Princeton, NJ, USA. He was a Founding Member of Snap Research, Los Angeles, CA, USA. He is currently a Visiting Professor with the School of Computer Science and Technology, University of Science and Technology of China. His research interests include computer vision and machine learning.

Dr. Wang has served as the Area Chair for CVPR and WACV. He served as the Chair for computer vision from 2015 to 2017.



Xinbo Gao (Senior Member, IEEE) received the B.Eng., M.Sc., and Ph.D. degrees in electronic engineering, signal and information processing from Xidian University, Xi'an, China, in 1994, 1997, and 1999, respectively.

From 1997 to 1998, he was a Research Fellow with the Department of Computer Science, Shizuoka University, Shizuoka, Japan. From 2000 to 2001, he was a Post-Doctoral Research Fellow with the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong. Since 2001, he has been with the School of Electronic Engineering, Xidian University. He is also a Cheung Kong Professor of the Ministry of Education of China, a Professor of pattern recognition and intelligent system with Xidian University, and a Professor of computer science and technology with the Chongqing University of Posts and Telecommunications, Chongqing, China. He has authored six books and around 300 technical articles in refereed journals and proceedings. His research interests include image processing, computer vision, multimedia analysis, machine learning, and pattern recognition.

Dr. Gao is a fellow of the Institute of Engineering and Technology and the Chinese Institute of Electronics. He served as the general chair/co-chair, the program committee chair/co-chair, or a PC member for around 30 major international conferences.

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